

# VIVEK PREETHAM MEKALA

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## PROFESSIONAL SUMMARY

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Motivated Computer Science undergraduate specializing in Data Science & Analytics at IIIT Nagpur, with strong Python skills and hands-on experience building production-grade AI/ML applications. Proven track record of developing NLP pipelines, deep learning models, and full-stack AI platforms with measurable accuracy improvements. Eager to contribute to enterprise-grade Generative AI and Agentic AI solutions at InteligenAI, with a keen interest in RAG systems, LLM integration, and scalable software development.

## EDUCATION

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**Indian Institute of Information Technology, Nagpur**

*B.Tech in Computer Science & Engineering (Data Science and Analytics)*

Nagpur, India

Aug 2023 – June 2027

## TECHNICAL SKILLS

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**Languages:** Python, C, C++, JavaScript

**AI / ML & Data Science:** TensorFlow, PyTorch, Keras, Scikit-learn, NLP, Deep Learning, Machine Learning, NumPy, Pandas, Matplotlib, Seaborn, OpenCV

**Frameworks & APIs:** Flask, Node.js, Express.js, Streamlit, REST APIs

**Databases:** MySQL, MongoDB

**Tools & Platforms:** Power BI, Tableau, Advanced Excel, Git

**Coursework:** DSA, OS, OOP, DBMS, CN, ML, DL, NLP, Computer Vision, Big Data

## PROJECTS

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**ByteBrain – AI-Powered Personalized Learning Platform** | *Flutter, Node.js, MongoDB, NLP, LLM* GitHub

- Engineered an end-to-end AI-powered learning platform adapting educational content based on learner preferences and proficiency levels, integrating an LLM-backed chatbot for intelligent content generation.
- Designed a modular application architecture combining retrieval strategies and candidate reranking, achieving up to **95% accuracy** with low-latency responses — directly analogous to RAG system design.
- Implemented analytics-driven personalization using NLP pipelines to process and serve contextually relevant learning material at scale.

**Twitter Sentiment Analysis System** | *Python, NLP, BiLSTM, TensorFlow*

- Developed a multi-class sentiment analysis solution leveraging BiLSTM with attention mechanism, improving classification accuracy from **92% to 95%** on a large-scale tweet corpus.
- Architected a comprehensive NLP preprocessing pipeline with exploratory data analysis to handle noisy social-media text, reducing neutral-class misclassification by approximately **15%**.
- Enhanced model interpretability by incorporating word-level attention, demonstrating ability to build and evaluate production-ready deep learning models.

**Movie Recommendation System** | *Python, NLP, TF-IDF, Streamlit*

GitHub

- Built a content-based movie recommendation engine using TF-IDF vectorization and cosine similarity, deployed as a real-time Streamlit web application integrated with an external movie metadata API.
- Constructed an efficient NLP preprocessing workflow and generated similarity matrices to support low-latency recommendations, demonstrating practical API integration and web deployment skills.

## LEADERSHIP & ACTIVITIES

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**Senior Design Member, Abhivyakti & Design Lead, Gamers Night** — Led visual branding, UI visuals, banners, and on-stage graphics for college-wide events.

**Lead – Visual Arts, Design Club “Strokes”** — Directed creative campaigns, illustrations, and design coordination for the club.